

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

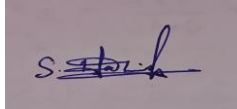
SHORT ANSWER TEST

Code of Conduct

I S. Harish (192511119) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

07/10/2026

CO1- AT3- SHORT ANSWER
TEST

S. Harish

Wednesday

CSA1520 - Cloud Computing
and BigData Analytics.

192511119

- 1) Define cloud computing and explain any four characteristics that make it different from traditional on premise computing.
- 2) Trace the evolution of cloud computing from hardware evolution to internet software evolution and show how these changes enabled modern cloud platforms.
- 3) Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as service model.
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- 5) Differentiate IaaS, PaaS, SaaS, XaaS with one real world example for each model.

1) cloud computing is the delivery of computing resources such as servers, storage devices, networking, software and processing power through the internet.

Characteristics:

- ★ on demand self service.
- ★ Broad network access
- ★ Resource pooling.
- ★ Rapid Elasticity.
- ★ Measured service.
- ★ High availability.

Elasticity allows resources to increase or decrease according to workload, while scalability supports handling growing demands. It supports technology such as virtual machines, hypervisors, virtual servers etc., virtual machines allows multiple virtual environments to run on the same physical hardware, while hypervisors manages these virtual machines. The cloud provider supplies and manages services.

2) Evolution of cloud Computing :

Mainframe Computing:

- ★ It came into existence in 1951.
- ★ It is highly powerful and reliable.
- ★ It is responsible for handling large data such as massive input-output operations.
- ★ It has no fault tolerance.
- ★ It is very expensive.

Distributed Computing:

- ★ It is a composition of multiple independent systems. But all of them are depicted as a single entity to the users.
- ★ The purpose of this is to share the resources and use them effectively.
- ★ It possesses characteristics such as concurrency, scalability, continuous availability.

Cluster Computing:

- ★ It is the process of grouping multiple independent systems.

Grid Computing:

- ★ It was introduced in 1990s.
- ★ Different systems located at different geographical locations are connected via internet.
- ★ The disadvantage is low availability of high bandwidth connections.

Utility Computing:

- ★ Introduced in Late 1990s - 2000.
- ★ utility computing is a computing model that defines service provisioning techniques for services such as computing along with various other services.

3) server virtualization is the technique of dividing one physical server into multiple virtual servers using hypervisors. Each VM can its own operating system and application.

- ★ Improves resource utilization.
- ★ Reduces hardware and operating cost.
- ★ Makes resource allocation faster and more reliable.